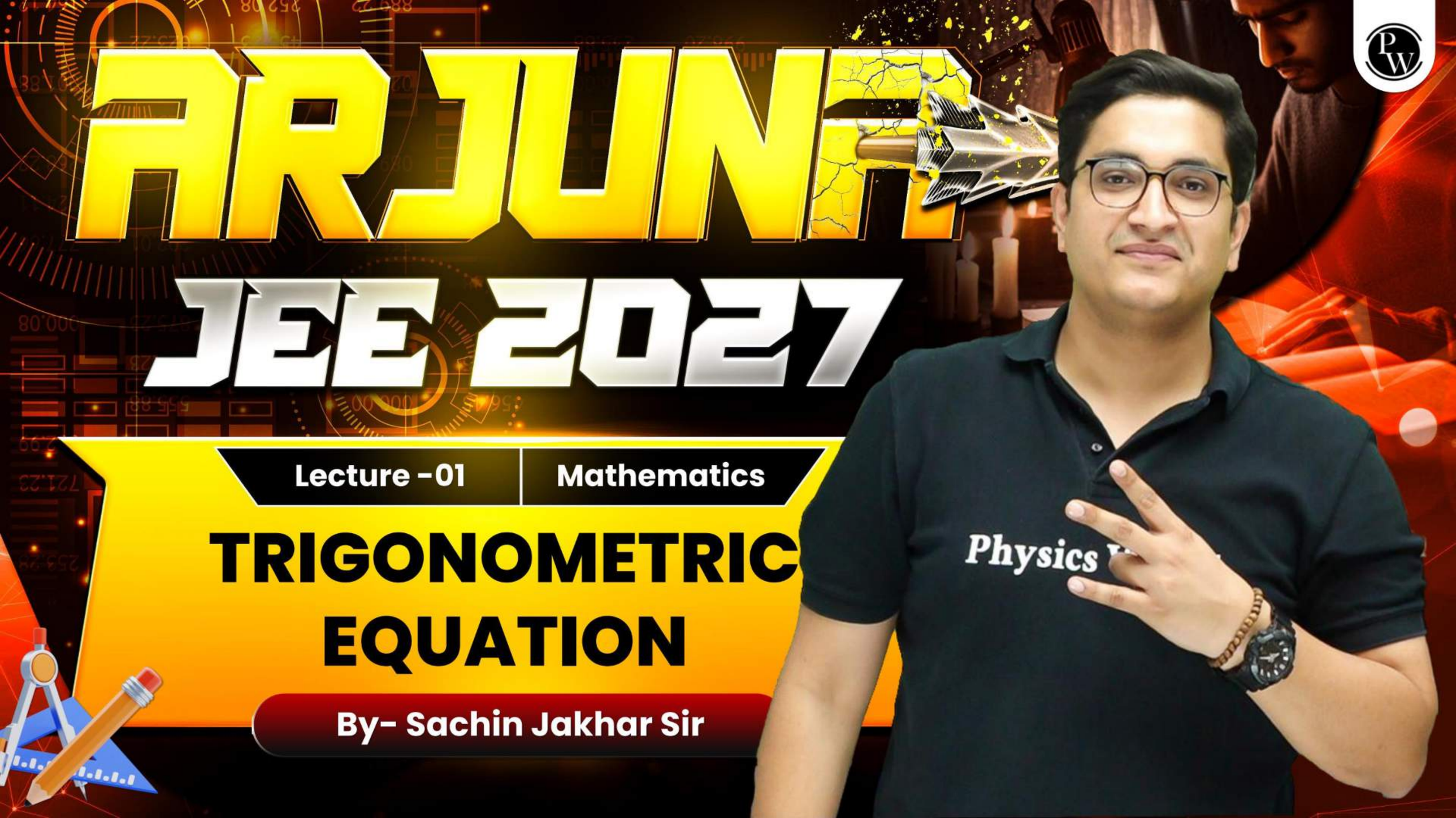
ARJUNA JEE 2027

Lecture -01 | Mathematics

TRIGONOMETRIC EQUATION

By- Sachin Jakhar Sir



Topics *to be covered*

A **Range of Trigonometric Functions**

B **Basics of Trigonometric Equations**

C **Question Practice**

D



H.W. Discussion

If $A = \theta, B = \theta + \frac{\pi}{3}$ & $C = \theta - \frac{\pi}{3}$, then show that :

(i) $\tan\theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta - \frac{\pi}{3}\right) = 3\tan 3\theta$

(ii) $\tan\theta \tan\left(\theta + \frac{\pi}{3}\right) + \tan\theta \tan\left(\theta - \frac{\pi}{3}\right) + \tan\left(\theta + \frac{\pi}{3}\right)\tan\left(\theta - \frac{\pi}{3}\right) = -3$

(iii) $\tan\theta \tan\left(\theta + \frac{\pi}{3}\right)\tan\left(\theta - \frac{\pi}{3}\right) = -\tan 3\theta$

(i) $\tan\theta + \tan(\overset{A}{\theta} + \overset{B}{60^\circ}) + \tan(\overset{A}{\theta} - \overset{B}{60^\circ})$

$\frac{\tan\theta + \sqrt{3}}{1 - \tan\theta \cdot \sqrt{3}} + \frac{\tan\theta - \sqrt{3}}{1 + \tan\theta \cdot \sqrt{3}}$

$t\theta + \sqrt{3} + \sqrt{3}t^2\theta + 3t\theta$

(iii) $-\tan\theta \cdot \tan(60^\circ + \theta)\tan(60^\circ - \theta)$

$-\tan 3\theta$

$$\tan\theta + \tan(\overset{A}{\theta}+\overset{B}{60^\circ}) + \tan(\overset{A}{\theta}-\overset{B}{60^\circ})$$

$$\frac{\tan\theta+\sqrt{3}}{1-\tan\theta\cdot\sqrt{3}} + \frac{\tan\theta-\sqrt{3}}{1+\tan\theta\cdot\sqrt{3}}$$

$$\tan\theta + \frac{t\theta+\sqrt{3}+\sqrt{3}t^2\theta+3t\theta+t\theta-\sqrt{3}-\sqrt{3}t^2\theta+3\tan\theta}{1^2-(\tan\theta\sqrt{3})^2}$$

$$\tan\theta + \frac{8\tan\theta}{1-3\tan^2\theta} = \frac{\tan\theta-3\tan^3\theta+8\tan\theta}{1-3\tan^2\theta} = 3\left(\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right)$$

$$= 3\tan 3\theta$$

If $A = \theta, B = \theta + \frac{\pi}{3}$ & $C = \theta - \frac{\pi}{3}$, then show that :

(i) $\tan\theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta - \frac{\pi}{3}\right) = 3\tan 3\theta$

(ii) $\tan\theta \tan\left(\theta + \frac{\pi}{3}\right) + \tan\theta \tan\left(\theta - \frac{\pi}{3}\right) + \tan\left(\theta + \frac{\pi}{3}\right)\tan\left(\theta - \frac{\pi}{3}\right) = -3$

(iii) $\tan\theta \tan\left(\theta + \frac{\pi}{3}\right)\tan\left(\theta - \frac{\pi}{3}\right) = -\tan 3\theta$

(ii) $A = \theta$, $\theta + 60^\circ = B$ & $\theta - 60^\circ = C$.

$\tan(\quad)$ $A + B + C = 3\theta$.

$\tan(A+B+C) = \tan 3\theta$.

$\frac{S_1 - S_3}{1 - S_2} = \tan 3\theta$.

$\frac{(i) - (iii)}{1 - (ii)} = \tan 3\theta$.

$4 = \frac{4\tan 3\theta}{\tan 3\theta} = \frac{3\tan 3\theta - (-\tan 3\theta)}{\tan 3\theta} = 1 - (ii)$

$1 - (ii) = 4$

$-3 = (ii)$

RANGE OF TRIGONOMETRIC FUNCTIONS

Type-01: Based on Range of Standard Trigonometric Functions

$\sin x \in [-1, 1]$

$\cos x \in [-1, 1]$

$\tan x \in R$

$\cot x \in R$

$\sec x \in (-\infty, -1] \cup [1, \infty)$

$\operatorname{cosec} x \in (-\infty, -1] \cup [1, \infty)$

RAR

Question

Find range of following functions :

(i) $y = 3\sin x - 5$

$\sin x \in [-1, 1]$

$2\sin x \in [-3, 3]$

$2\sin x - 5 \in \boxed{[-8, -2]}$

(ii) $y = \frac{1}{5\cos x + 2}$

(iii) $y = \sqrt{2\sec x - 1}$

$\cos x \in [-1, 1]$

$5\cos x \in [-5, 5]$

$5\cos x + 2 \in [-3, 7]$

$\frac{1}{5\cos x + 2} \in \left(-\infty, \frac{-1}{3}\right] \cup \left[\frac{1}{7}, \infty\right)$

Question

Page No.48

Find range of following functions :

(iv) $y = \sin^4 x + \cos^4 x$

(v) $y = 7\sin^2 x + 2\cos^2 x + 3$

$y = 5\sin^2 x + \underbrace{2\sin^2 x + 2\cos^2 x}_{2} + 3$

$= 5\sin^2 x + 5 \in [5, 10]$

$\sin x \in [-1, 1]$

$\sin^2 x \in [0, 1]$

$5\sin^2 x \in [0, 5]$

(vi) $y = \dfrac{\sin 3x}{\sin x}$

$y = \dfrac{3\sin x - 4\sin^3 x}{\sin x}$

$y = 3 - 4\sin^2 x$

$\sin^2 x \in [0, 1]$

$-4\sin^2 x \in [-4, 0]$

$3 - 4\sin^2 x \in [-1, 3]$ ✗

(vi) $y = \dfrac{2\sin x - 4\sin^3 x}{\sin x}$

$\Rightarrow y = 3 - 4\sin^2 x$.

$\sin^2 x \in [0,1]$

$\Rightarrow -4\sin^2 x \in [-4,0]$

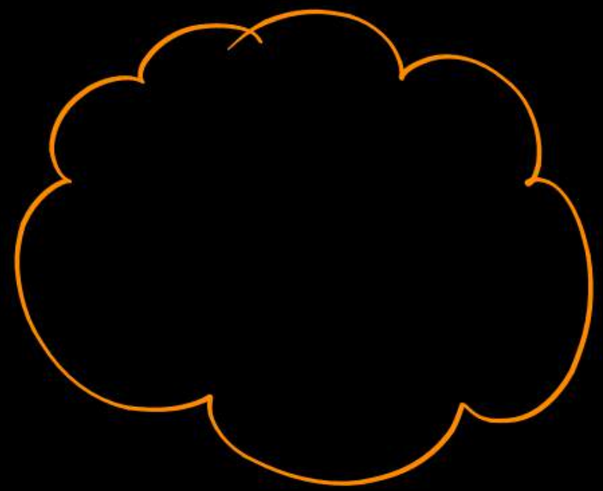
$\Rightarrow 3 - 4\sin^2 x \in [-1,3]$ ✗

$\sin^2 x \in (0,1]$

$\Rightarrow -4\sin^2 x \in [-4,0)$

$\Rightarrow 3 - 4\sin^2 x \in [-1,3)$ ✓

$y = \dfrac{\sin 3x}{\sin x}$ → domain $(\sin x) \neq 0$.



METHOD TO COMPLETE PERFECT SQUARE

STEPS	EXAMPLE $(3x^2+5x+4)$
S-1: Make coefficient of x^2 equals to 1. ✓	$3\left(x^2+\frac{5}{3}x+\frac{4}{3}\right)$
S-2: Now, coefficient of x is multiplied & divided by 2 ✓	$3\left(x^2+2\left(\frac{5}{6}\right)x+\frac{4}{3}\right)$
S-3: Add & Subtract, square of coefficient of $2x$.	$3\left(x^2+2\left(\frac{5}{6}\right)x+\left(\frac{5}{6}\right)^2-\left(\frac{5}{6}\right)^2+\frac{4}{3}\right)$
S-4: Complete square, simplify constant term and enjoy	$3\left(\left(x+\frac{5}{6}\right)^2+\frac{23}{36}\right)=3\left(x+\frac{5}{6}\right)^2+\frac{23}{12}$

RAPID FIRE ON COMPLETING PERFECT SQUARE

Show that :

(i) $x^2 - 4x + 9 = (x-2)^2 + 5$

(ii) $x^2 - 3x + 7 = \left(x - \frac{3}{2}\right)^2 + \frac{19}{4}$

} DIBY

(iii) $3x^2 + 3x + 1 = 3\left(x + \frac{1}{2}\right)^2 + \frac{1}{4}$

$3\left(x^2 + 2\left(\frac{1}{2}\right)x + \frac{1}{3}\right)$

$3\left(x^2 + 2\,\frac{1}{2}\,x + \frac{1}{4} - \frac{1}{4} + \frac{1}{3}\right)$

$3\left(\left(x+\frac{1}{2}\right)^2 + \frac{1}{12}\right) = 3\left(x+\frac{1}{2}\right)^2 + \frac{1}{4}$

(iv) $-2x^2 + 20x - 43 = 7 - 2(x-5)^2$ ✓

$-2\left(x^2 - 2\left(\frac{10}{2}\right)x + \frac{43}{2}\right)$ (5)

$-2\left(x^2 - 2(5)x + 25 - 25 + \frac{43}{2}\right) = -2\left((x-5)^2 - \frac{7}{2}\right)$

$= -2(x-5)^2 + 7.$

RANGE OF TRIGONOMETRIC FUNCTIONS

Type-02 : Range by completing perfect square

Question

(i) $y = \cos^2 x - 4\cos x + 13$

(ii) $y = \cos 2x - 5\sin x + 1$

(iii) $y = \sin^2 x + \cos^4 x$

(ii) $y = 1 - 2\sin^2 x - 5\sin x + 1$.

put $\cos x = t$

$y = t^2 - 4t + 13$.

$y = t^2 - 2 \times 2t + 4 - 4 + 13$

$y = (t-2)^2 + 9$.

$y = (\cos x - 2)^2 + 9$.

$\cos x \in [-1, 1]$

$\cos x - 2 \in [-3, -1]$

$(\cos x - 2)^2 \in [1, 9]$

$(\cos x - 2)^2 + 9 \in [10, 18]$

(ii) $y = 1 - 2\sin^2 x - 5\sin x + 1$.

$\sin x = t$.

$y = -2t^2 - 5t + 2$

$= -2\left(t^2 + 2\cdot\frac{5}{2\times 2}t - 1\right)$

$= -2\left(t^2 + 2\cdot\frac{5}{2\times 2}t + \frac{25}{16} - \frac{25}{16} - 1\right)$

$y = -2\left(t + \frac{5}{4}\right)^2 + \frac{41}{8}$

$y = -2\left(\sin x + \frac{5}{4}\right)^2 + \frac{41}{8}$

$\sin x \in [-1, 1]$

$\sin x + \frac{5}{4} \in \left[\frac{1}{4}, \frac{9}{4}\right]$

$\left(\sin x + \frac{5}{4}\right)^2 \in \left[\frac{1}{16}, \frac{81}{16}\right]$

$-2\left(\sin x + \frac{5}{4}\right)^2 \in \left[-\frac{81}{8}, -\frac{1}{8}\right]$

$-2\left(\sin x + \frac{5}{4}\right)^2 + \frac{41}{8} \in \boxed{[-5, 5]}$

RANGE OF TRIGONOMETRIC FUNCTIONS

Type-02 : Range by completing perfect square

Question

(i) $y = \cos^2 x - 4\cos x + 13$

(ii) $y = \cos 2x - 5\sin x + 1$

(iii) $y = \sin^2 x + \cos^4 x$

DIBY - (X) ✓

$y = 1 - \cos^2 x + \cos^4 x$

$\cos^2 x = t$

$y = t^2 - t + 1$

RANGE OF TRIGONOMETRIC FUNCTIONS

Type-03 : Form : $f(x) = a\cos x \pm b\sin x$



$$\text{Range} \in \left[-\sqrt{a^2+b^2},\ \sqrt{a^2+b^2}\right]$$

RAR

before applying this :

(i) angle should be same.

(ii) powers of sin & cos should '1'.

(iii) x (angle) $\in R$

RAM

$f(x) = a\cos x \pm b\sin x$

multiply & divide by $\sqrt{a^2+b^2}$

$$f(x) = \sqrt{a^2+b^2}\left(\frac{a\cos x \pm b\sin x}{\sqrt{a^2+b^2}}\right)$$

$$f(x) = \sqrt{a^2+b^2}\left(\frac{a}{\sqrt{a^2+b^2}}\cos x \pm \frac{b}{\sqrt{a^2+b^2}}\sin x\right)$$

$\frac{a}{\sqrt{a^2+b^2}} \rightarrow \sin\theta$, $\frac{b}{\sqrt{a^2+b^2}} \rightarrow \cos\theta$

$$f(x) = \sqrt{a^2+b^2}\left(\sin\theta\cos x \pm \cos\theta\sin x\right)$$

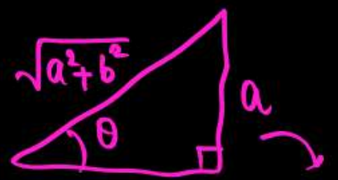
$$\Downarrow$$

$$f(x) = \sqrt{a^2+b^2}\sin(\theta \pm x) \in \left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$$

$\sin(\theta \pm x) \rightarrow [-1, 1]$

$\sqrt{a^2+b^2} \times [-1, 1]$

Say: $\sin\theta = \frac{a}{\sqrt{a^2+b^2}} = \frac{P}{H}$



$\cos\theta = \frac{\pm b}{\sqrt{a^2+b^2}}$

Proof:

$$f(x) = a\cos x \pm b\sin x$$

$$\boxed{(a_1b_1 + a_2b_2)^2 \le (a_1^2 + a_2^2)(b_1^2 + b_2^2)} \quad (\text{Cauchy Ineq.})$$

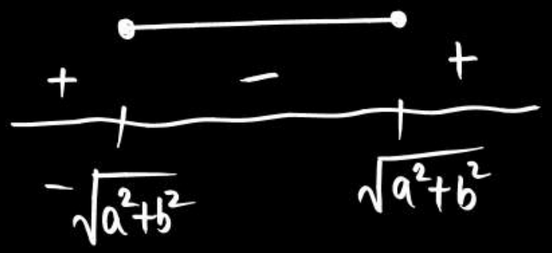
$$(a\cos x \pm b\sin x)^2 \le (a^2+b^2)(\underbrace{\cos^2 x + \sin^2 x}_{=1})$$

(a_1 = a, b_1 = $\cos x$, a_2 = b, b_2 = $\sin x$)

$$(f(x))^2 \le (a^2+b^2)$$

$$(f(x))^2 - (a^2+b^2) \le 0$$

$$\left(f(x) - \sqrt{a^2+b^2}\right)\left(f(x) + \sqrt{a^2+b^2}\right) \le 0$$



Ques:-

(i) $y = 1\sin x + 1\cos x$ → $a = b = 1$, $\sqrt{a^2+b^2} = \sqrt{2}$.

→ Range $\in [-\sqrt{2}, \sqrt{2}]$ ✓

(ii) $y = 1\sin x - 1\cos x$

→ Range $\in [-\sqrt{2}, \sqrt{2}]$

(iii) $y = 2\cos x - 1\sin x$

$a = 2$, $b = 1$.

$\sqrt{a^2+b^2} = \sqrt{5}$.

→ Range $\in [-\sqrt{5}, \sqrt{5}]$

$y = 1\sin x + 1\cos x$

→ Max. $= 2$

→ Min. $= -2$.

→ Range $\in [-2, 2]$ ✗

Question [JEE-1988]

"Notes"

The value of the expression $\sqrt{3}\text{cosec}20° - \sec 20°$ is equal to

A. 2

B. 2sin20°/sin40°

C. 4

D. 4sin20°/sin40°

$$\frac{\sqrt{3}}{\sin 20°} - \frac{1}{\cos 20°}$$

$$\frac{\sqrt{3}\cos 20° - \sin 20°}{\frac{1}{2}(2\sin 20° \cos 20°)}$$

$$\frac{2\sin 40°}{\frac{1}{2}\sin 40°} = 4$$

$a\cos\theta \pm b\sin\theta$

$\sqrt{3}\cos 20° - 1\sin 20°$

$\times \div$ by $\sqrt{(\sqrt{3})^2 + 1^2} = 2$

$$2\left(\frac{\sqrt{3}}{2}\cos 20° - \frac{1}{2}\sin 20°\right)$$

$\frac{\sqrt{3}}{2} = \cos 30°$, $\frac{1}{2} = \sin 30°$

$2(\cos 30°\cos 20° - \sin 30°\sin 20°)$

$\Downarrow$

$2\cos(30° + 20°)$

$2(\sin 60°\cos 20° - \cos 60°\sin 20°)$

$\Downarrow$

$2\sin(60° - 20°)$

same

Question

Find the exact value of

(a) $\sqrt{3}\text{cosec}20^\circ - \sec 20^\circ$

(b) $\text{cosec}10^\circ - \sqrt{3}\sec 10^\circ$



Question

Page No.48

Find the maximum & minimum value of :

(i) $y = 3\cos\theta - 4\sin\theta + 6 \in [1, 11]$

$3\cos\theta - 4\sin\theta \in [-5, 5]$

$\sqrt{3^2+4^2} = 5.$

(ii) $y = 3\cos\left(\overset{A}{x} + \overset{B}{\frac{\pi}{3}}\right) + 5\cos x + 3$

(iii) $y = 5\sin x \cos x - \sin^2 x + 2$

$y = 3\left(\cos x \cdot \frac{1}{2} - \sin x \cdot \frac{\sqrt{3}}{2}\right) + 5\cos x + 3.$

$= \frac{3}{2}\cos x - \frac{3\sqrt{3}}{2}\sin x + 5\cos x + 3$

$= \underbrace{\frac{13}{2}}_{a}\cos x - \underbrace{\frac{3\sqrt{3}}{2}}_{b}\sin x + 3.$

$\sqrt{a^2+b^2} = \sqrt{\frac{169}{4} + \frac{27}{4}} = \sqrt{49} = 7.$

$[-7, 7] + 3 \equiv \boxed{[-4, 10]}$

Question

Page No.48

Find the maximum & minimum value of :

(i) $y = 3\cos\theta - 4\sin\theta + 6$

(ii) $y = 3\cos\left(x + \frac{\pi}{3}\right) + 5\cos x + 3$

(iii) $y = 5\sin x \cos x - \sin^2 x + 2$

$\cos 2x = 1 - 2\sin^2 x$

$\sin^2 x = \frac{1 - \cos 2x}{2}$

$y = 5\left(\frac{1}{2}\sin 2x\right) - \left(\frac{1}{2} - \frac{\cos 2x}{2}\right) + 2$

$y = \frac{5}{2}\sin 2x + \frac{1}{2}\cos 2x + \frac{3}{2}$

$\left[-\sqrt{\frac{25}{4} + \frac{1}{4}}, \sqrt{\frac{25}{4} + \frac{1}{4}}\right]$

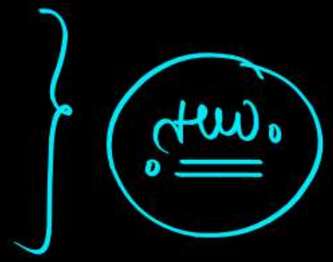
Question

Page No.49

Find the maximum & minimum value of :

(iii) $y = \log_2\left(\frac{3\sin x - 4\cos x + 15}{10}\right)$

(iv) $y = 27^{\cos 3x} \cdot 81^{\sin 3x}$



Question [IIT-JEE-2010]

Find the maximum & minimum value of :

$$y = \frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$$



RANGE OF TRIGONOMETRIC FUNCTIONS

Type-04 : *Special Form* (Using Reciprocal Relationship)

(i) $a^2 \tan^2\theta + b^2 \cot^2\theta \geq 2ab$

(ii) $a^2 \sec^2\theta + b^2 \operatorname{cosec}^2\theta \geq (a+b)^2$

} # RAR

(where a & b are +ve real numbers)

$a^2(\sec^2\theta - 1) + b^2(\operatorname{cosec}^2\theta - 1) \geq 2ab$

$a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta \geq a^2+b^2+2ab$

$\downarrow$

$(a+b)^2$

$y = a^2 \tan^2\theta + b^2 \cot^2\theta$

$$AM \geq GM$$

$$\frac{a^2 \tan^2\theta + b^2 \cot^2\theta}{2} \geq \left(a^2 \tan^2\theta \, b^2 \cot^2\theta\right)^{\frac{1}{2}}$$

$$a^2 \tan^2\theta + b^2 \cot^2\theta \geq 2ab$$

Question

The minimum value of $3\tan^2\theta + 12\cot^2\theta$ is

A 6

B 15

C 24

D 12

$a^2 = 3 \rightarrow a = \sqrt{3}$

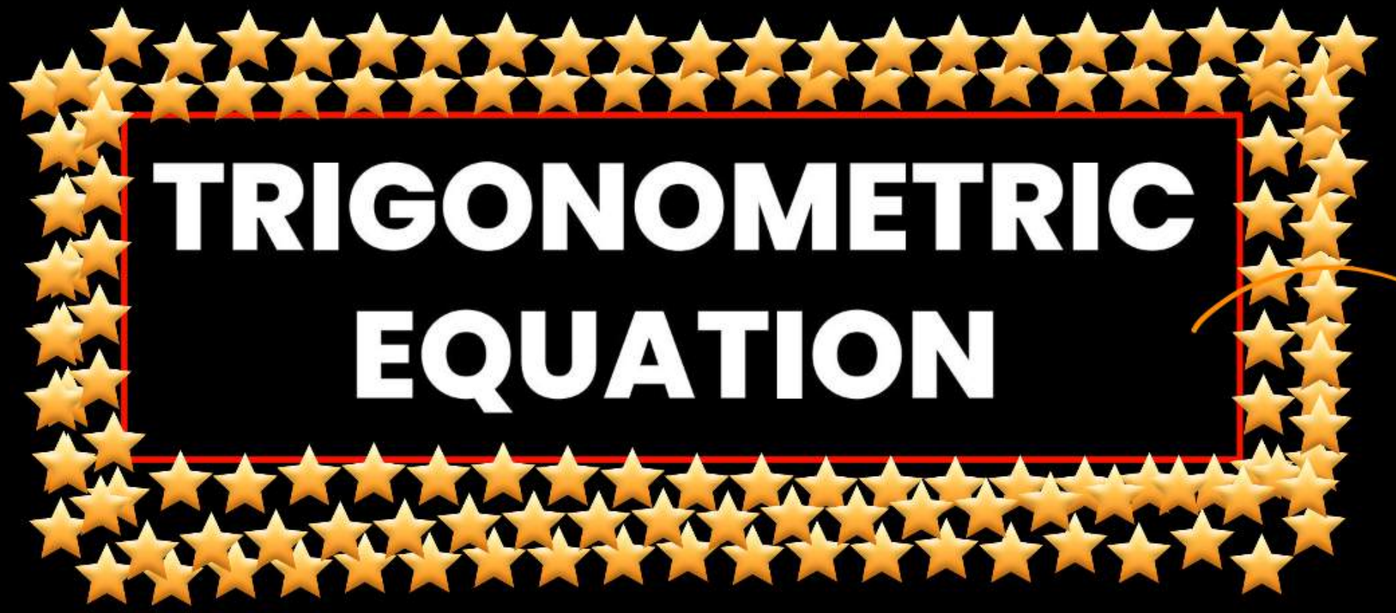
$b^2 = 12 \rightarrow b = 2\sqrt{3}$.

min. value $= 2ab$.

$= 2\sqrt{3} \times 2\sqrt{3}$.

$= 12$.

$a^2\tan^2\theta + b^2\cot^2\theta \geq 2ab$



1 Ques^n - - Mains
↳ Adv.

THINK ANGLE CORRESPONDING TO VALUE

I + II

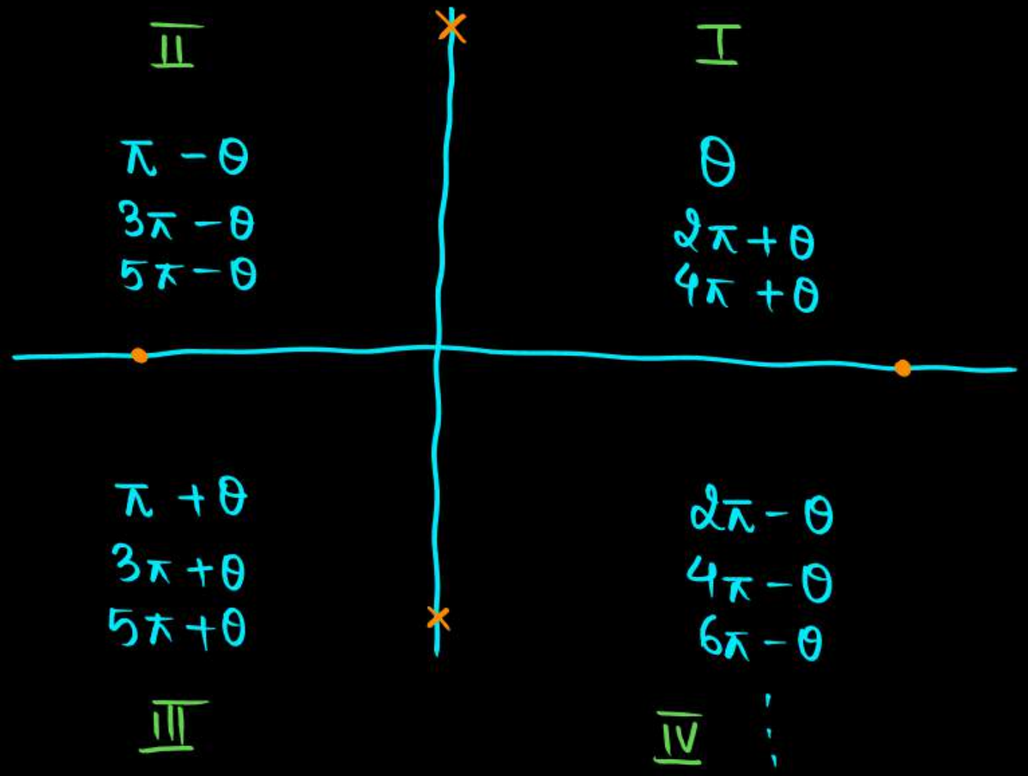
$\sin\theta = +\frac{1}{2}$

$\theta = 30°$ ... ??

S-I: Value के corresp. (acute angle)

S-II: Sign देखो & decide Quad.

S-III: Rotate that angle in required Quad.



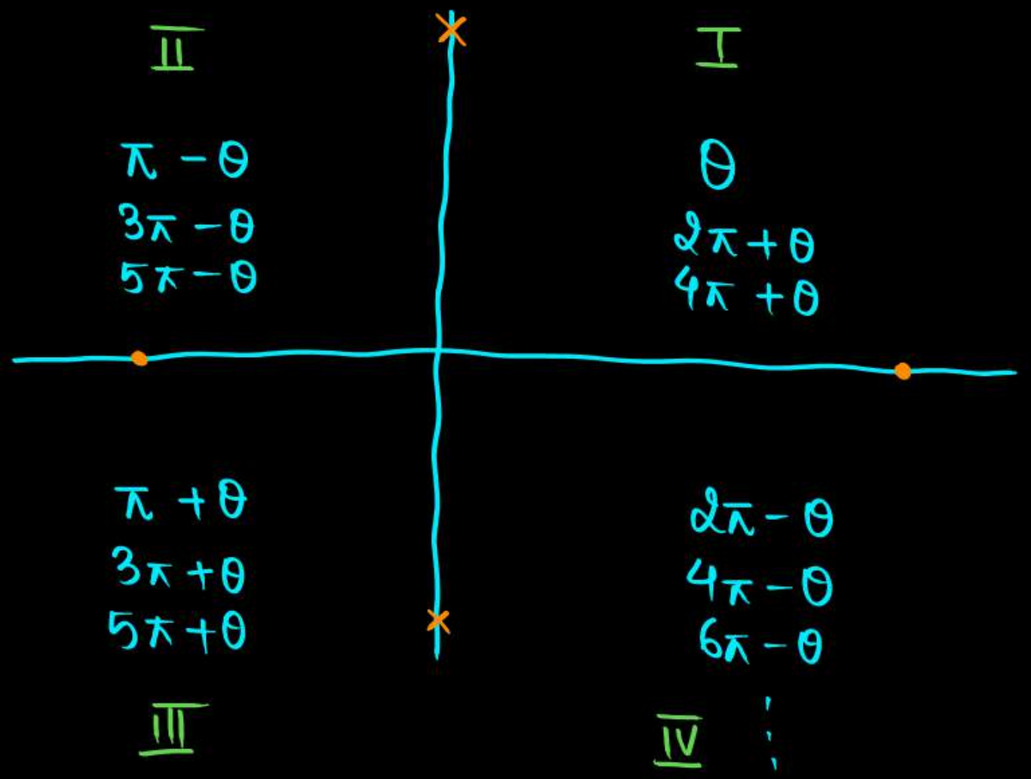
I + II

$\sin x = \frac{1}{2}$

$\theta = 30^\circ$ or $\frac{\pi}{6}$

\# $\frac{\pi}{6}, 2\pi + \frac{\pi}{6}, 4\pi + \frac{\pi}{6}$ ----

$\pi - \frac{\pi}{6}, 3\pi - \frac{\pi}{6}, 5\pi - \frac{\pi}{6}$ -----

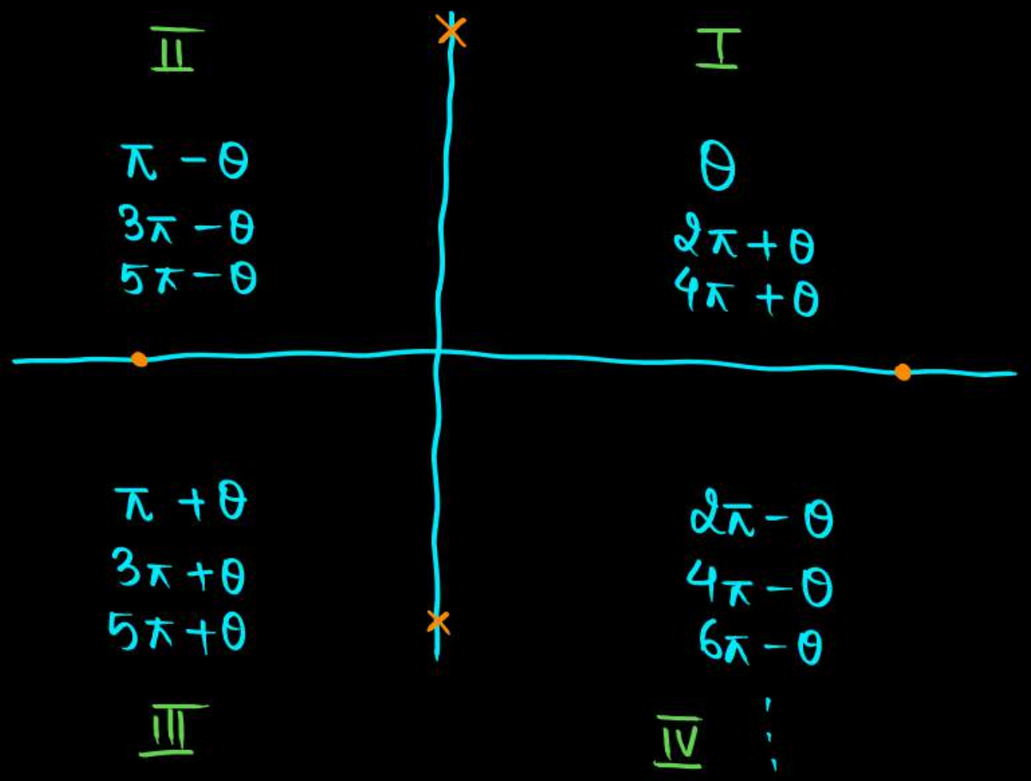


III OR IV

$\sin x = \ominus \frac{1}{2}$

$\theta = 30^\circ$ OR $\frac{\pi}{6}$

$$x = \begin{cases} \pi + \frac{\pi}{6}, 3\pi + \frac{\pi}{6}, 5\pi + \frac{\pi}{6} \ldots \\ 2\pi - \frac{\pi}{6}, 4\pi - \frac{\pi}{6}, 6\pi - \frac{\pi}{6} \ldots \end{cases}$$



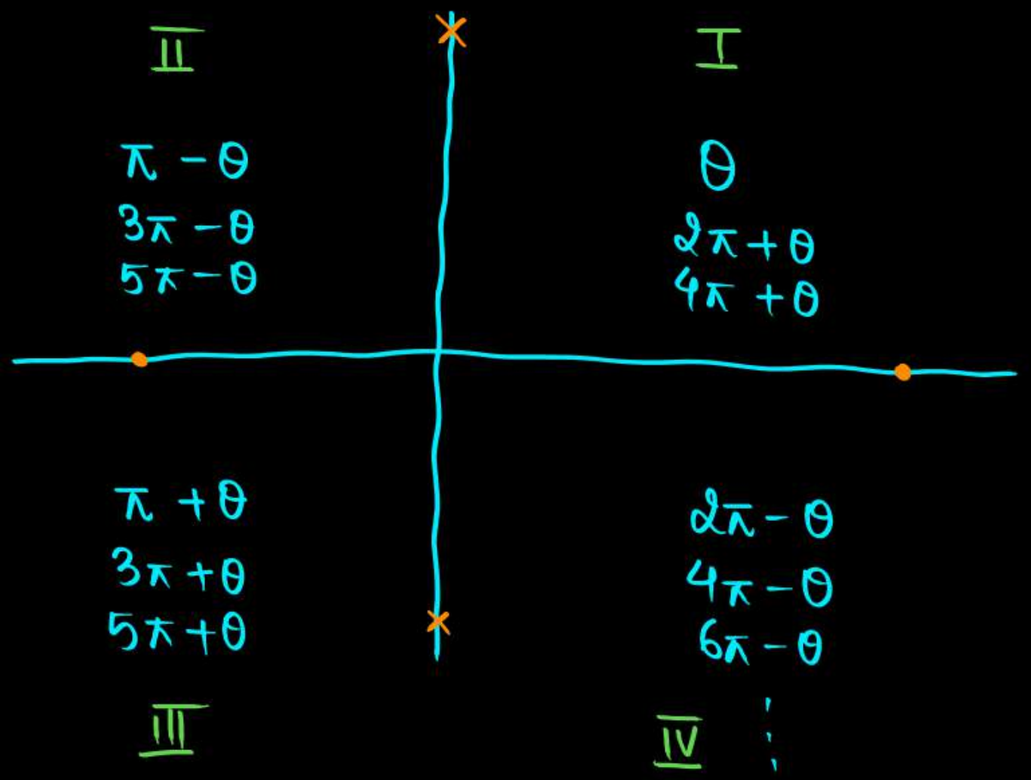
II OR III

$\cos x = \ominus \frac{1}{\sqrt{2}}$

$\theta = \frac{\pi}{4}$

$\pi - \frac{\pi}{4}$, $3\pi - \frac{\pi}{4}$, $5\pi - \frac{\pi}{4}$ . . . .

$\pi + \frac{\pi}{4}$, $3\pi + \frac{\pi}{4}$, $5\pi + \frac{\pi}{4}$ . . . .



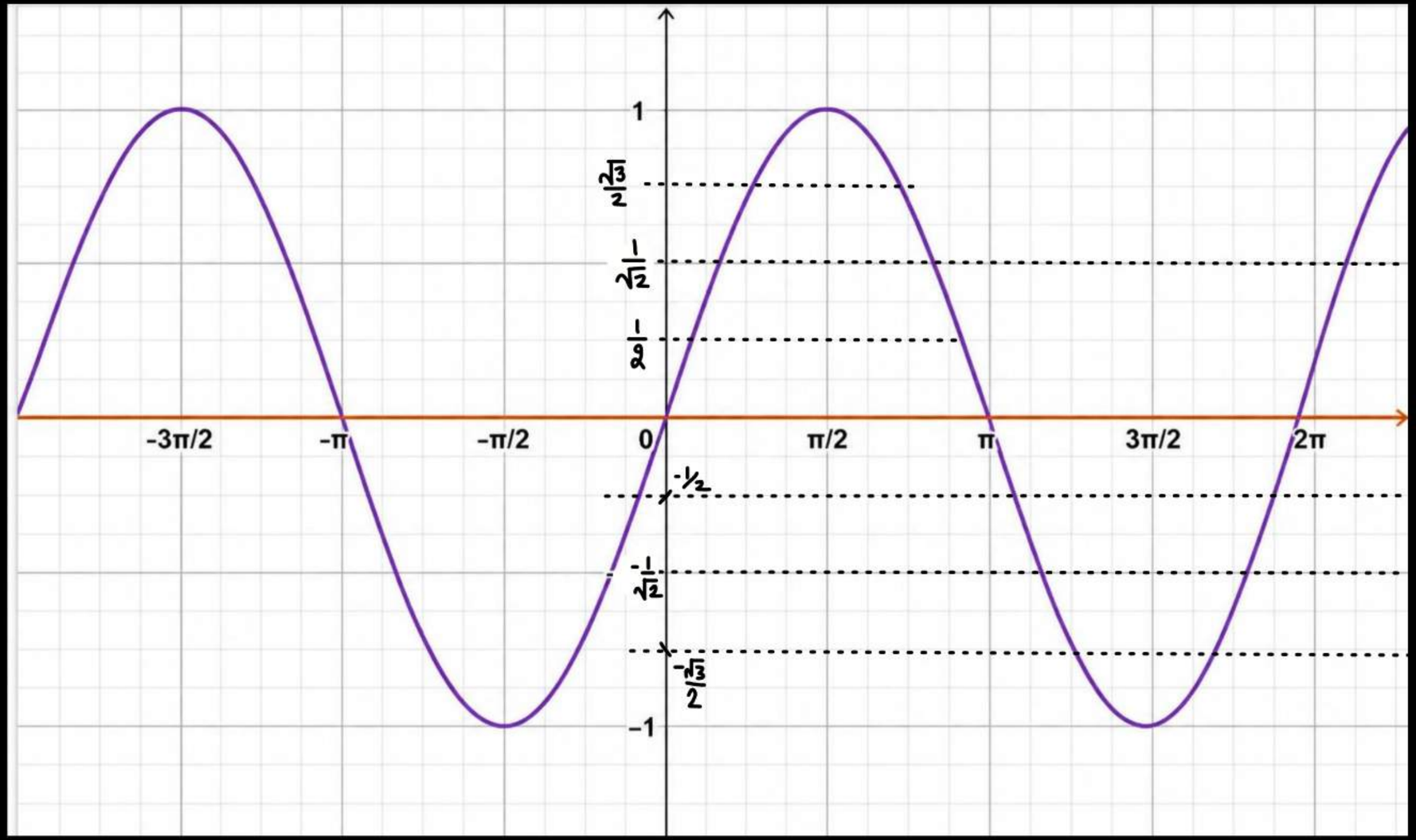
$\tan x = \sqrt{3}$ (I or III)

$\theta = \frac{\pi}{3}$

$\frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 4\pi + \frac{\pi}{3}$ - - - -

$\pi + \frac{\pi}{3}, 3\pi + \frac{\pi}{3}, 5\pi + \frac{\pi}{3}$ - - - -

THINK ANGLE CORRESPONDING TO VALUE



RAPID FIRE DIBY'S

DIBY-A

The expression $\frac{\cos^3\alpha - \cos 3\alpha}{\cos\alpha} + \frac{\sin^3\alpha + \sin 3\alpha}{\sin\alpha}$ is equal to:

$(4\cos^3\alpha - 3\cos\alpha)$ $\quad$ $3\sin\alpha - 4\sin^3\alpha$

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : C]

DIBY-B

Express $\cos 5A$ in terms of $\cos A$

(A) $16\cos^5 A - 20\cos^3 A + 5\cos A$ (B) $16\cos^5 A - 12\cos^3 A + 3\cos A$

(C) $16\cos^5 A - 20\cos^3 A + 3\cos A$ (D) $32\cos^2 A - 32\cos^3 A + 6\cos A$

[Ans. : A]

DIBY-C

The value of the product $\cos 36^\circ . \cos 72^\circ$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{\sqrt{5}}{4}$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-D

Show that: (i) $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$

(ii) $\sin 75^\circ - \sin 15^\circ = \cos 105^\circ + \cos 15^\circ$

(iii) $\sin((n+1)A)\sin((n-1)A) + \cos((n+1)A)\cos((n-1)A) = \cos 2A$

DIBY-E

Let $S = \sin\sqrt{2} - \sin\sqrt{3}$ and $C = \cos\sqrt{2} - \cos\sqrt{3}$ then which one is correct?

(A) $S > 0$ and $C > 0$ (B) $S > 0$ and $C < 0$

(C) $S < 0$ and $C > 0$ (D) $S < 0$ and $C < 0$

-ve

$2\sin\frac{\sqrt{2}-\sqrt{3}}{2}\cos\frac{\sqrt{2}+\sqrt{3}}{2}$

DIBY-F

The expression $\frac{\sec 8A - 1}{\sec 4A - 1}$ is equivalent to :

$\frac{\frac{1}{\cos 8A} - 1}{\frac{1}{\cos 4A} - 1} = \frac{1-\cos 8A}{1-\cos 4A} \times \frac{\cos 4A}{\cos 8A}$

$1-\cos 8A = 2\sin^2 4A$, $1-\cos 4A = 2\sin^2 2A$

$= \frac{2\sin 4A\cos 4A \cdot \sin 4A}{\cos 8A \cdot 2\sin^2 2A}$, $2\sin 4A\cos 4A = \sin 8A$

$= \frac{\sin 8A \cdot \sin 4A}{\cos 8A \cdot 2\sin^2 2A}$, $2\sin 2A\cos 2A = \sin 4A$

$= \frac{\sin 8A}{2\cos 8A} = \frac{1}{2}\tan 8A$

(A) $\frac{\tan 8A}{\tan 4A}$ (B) $\frac{\tan 8A}{\tan 2A}$ (C) $\frac{\tan 4A}{\tan -2A}$ (D) $\frac{\tan 2A}{\tan -8A}$

[Ans. : A]

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-G

What is the exact value of $\cos^2 48^\circ - \sin^2 12^\circ$

(A) $\frac{\sqrt{5}-1}{8}$ (B) $\frac{\sqrt{5}+1}{8}$ (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{5}-1}{4}$

[Ans. : B]

DIBY-H

$\frac{\pi}{10} = \theta$

$4\cos\theta - \frac{3}{\cos\theta} = \frac{4\cos^2\theta - 3}{\cos\theta} = \frac{4\cos^3\theta - 3\cos\theta}{\cos^2\theta}$

$\frac{\cos 3\theta}{\cos^2\theta}$

What is the exact value of $4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$

(A) 0 (B) 1 (C) $\sqrt{5}$ (D) $\frac{1}{2}$

[Ans. : A]

$A = 25^\circ,\ B = 50^\circ,\ C = 105^\circ \rightarrow A+B+C = 180^\circ$

DIBY-I

The value of the expression $\frac{\sin 50^\circ + \sin 100^\circ + \sin 210^\circ}{\sin 25^\circ \sin 50^\circ \sin 105^\circ}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

$\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A \sin B \sin C}$

[Ans. : D]

RAPID FIRE DIBY'S

DIBY-J

The value of the product $\sin\frac{\pi}{16}\cdot\sin\frac{3\pi}{16}\cdot\sin\frac{5\pi}{16}\cdot\sin\frac{7\pi}{16}$ is equal to:

(A) $\frac{1}{16}$ (B) $\frac{1}{16\sqrt{2}}$ (C) $\frac{1}{8\sqrt{2}}$ (D) $\frac{1}{8}$

[Ans. : C]

DIBY-K

Find the value of $\prod_{r=1}^{7}\sin(2r-1)\frac{\pi}{14}$



(A) $\frac{1}{64}$ (B) $\frac{1}{128}$ (C) $\frac{1}{256}$ (D) $\frac{1}{32}$

[Ans. : A]

DIBY-L

If $\sin\theta+\sin\phi=a$ and $\cos\theta+\cos\phi=b$, then the value of $\tan\left(\frac{\theta-\phi}{2}\right)$ is:

(A) $\pm\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$ (B) $\pm\sqrt{\frac{a^2+b^2}{4-a^2-b^2}}$ (C) $\pm\sqrt{\frac{4+a^2+b^2}{a^2+b^2}}$ (D) $\pm\sqrt{\frac{a^2+b^2}{4+a^2+b^2}}$

[Ans. : A]

RAPID FIRE DIBY'S

DIBY-M

Show that: $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0.$

DIBY-N

Show that: $\frac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)} = \cot B$

DIBY-O

If $x - y = \frac{\pi}{4}$ and $\cot x + \cot y = 2$. then the smallest positive angles x and y are given by the ordered pair $(x, y) =$

(A) $\left(\frac{7\pi}{12}, \frac{\pi}{3}\right)$ (B) $\left(\frac{5\pi}{12}, \frac{\pi}{6}\right)$ (C) $\left(\frac{\pi}{3}, \frac{\pi}{12}\right)$ (D) $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-P

If α, β and γ are the interior angles of a triangle, then the value of the expression $\frac{\cos\alpha}{\sin\beta\sin\gamma} + \frac{\cos\beta}{\sin\alpha\sin\gamma} + \frac{\cos\gamma}{\sin\alpha\sin\beta}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : B]

DIBY-Q

Find the values of (1) $\sin 7\frac{1}{2}^\circ$ (2) $\cos 7\frac{1}{2}^\circ$ (3) $\tan 11\frac{1}{4}^\circ$.

[Ans. : (i) $\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$, (ii) $\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}}$; (iii) $-(\sqrt{2}+1)+\sqrt{4+2\sqrt{2}}$]

DIBY-R

Show that: $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ = 1$

DO IT BY YOURSELF (DIBY)

DIBY-35

Value of : $\prod_{k=1}^{17} \sin\frac{k\pi}{18} = \frac{a}{2^{16}}$ then $a =$ _____.

[Ans. : 9]

DIBY-36

If $\frac{5\pi}{2} < x < 3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$

(B) $\cot\frac{x}{2}$

(C) $\tan\frac{x}{2}$

(D) $-\tan\frac{x}{2}$

[Ans. : (C)]

DIBY-37

If $A = 340^\circ$ then $2\sin\frac{A}{2}$ is identical to

(A) $\sqrt{1+\sin A} + \sqrt{1-\sin A}$

(B) $-\sqrt{1+\sin A} - \sqrt{1-\sin A}$

(C) $\sqrt{1+\sin A} - \sqrt{1-\sin A}$

(D) $-\sqrt{1+\sin A} + \sqrt{1-\sin A}$

[Ans. : (D)]

DIBY-38

Evaluate: $\cos^6\frac{\pi}{16} + \cos^6\frac{3\pi}{16} + \cos^6\frac{5\pi}{16} + \cos^6\frac{7\pi}{16}$

(A) $5/8$ (B) $5/4$ (C) $7/16$ (D) $5/32$

[Ans. : (B)]

DIBY-39

If $\frac{x}{\cos\theta} = \frac{y}{\cos\left(\theta+\frac{2\pi}{3}\right)} = \frac{z}{\cos\left(\theta-\frac{2\pi}{3}\right)}$, then $x+y+z$ is equal to:

(A) –1 (B) 1 (C) 0 (D) 2

[Ans. : (C)]

DIBY-40

Find the exact value of $\tan^2\frac{\pi}{16}+\tan^2\frac{3\pi}{16}+\tan^2\frac{5\pi}{16}+\tan^2\frac{7\pi}{16}$

[Ans. : 28]

DIBY-41

Prove that $\cos36^\circ\cos72^\circ\cos108^\circ\cos144^\circ = 1/16$.

DIBY-42

If $A+B+C=\pi$, prove that $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$.

DIBY-43

If $A+B+C=180^\circ$, prove that $\cos^2A+\cos^2B+\cos^2C=1-2\cos A\cos B\cos C$.

DIBY-44

In triangle ABC, prove that

$\sin(B+C-A)+\sin(C+A-B)+\sin(A+B-C)=4\sin A\sin B\sin C$.

DIBY-45

If $A+B+C=\pi/2$, show that

(a) $\sin^2 A+\sin^2 B+\sin^2 C=1-2\sin A\sin B\sin C$

(b) $\cos^2 A+\cos^2 B+\cos^2 C=2+2\sin A\sin B\sin C$

DIBY-46

If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$, then [IIT-JEE-2009]

(A) $\tan^2 x=\frac{2}{3}$

(B) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{1}{125}$

(C) $\tan^2 x=\frac{1}{3}$

(D) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{2}{125}$

[Ans. : A, B]

DIBY-47

For $\triangle ABC$, $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}-1=\lambda\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}\sin\frac{\pi-C}{4}$. Then $\lambda=$

[Ans. : 4]

DIBY-48

The value of $\sin^3 10^\circ+\sin^3 50^\circ-\sin^3 70^\circ$ is equal to

(A) $-\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{4}$ (D) $-\frac{3}{8}$

[Ans. : (D)]

DIBY-49

If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$ then the value of $\cot^2 x$ is equal to _____.

[Ans. : 3/2]

DIBY-50

The value of $\frac{\tan(\ln 6)-\tan(\ln 2)-\tan(\ln 3)}{\tan(\ln 6)\cdot\tan(\ln 2)\cdot\tan(\ln 3)}$ is equal to _____.

[Ans. : 1]

DIBY-51

The value of $\cos^3\left(\frac{\pi}{8}\right)\cdot\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\cdot\sin\left(\frac{3\pi}{8}\right)$ is [JEE Mains-2020]

(A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

[Ans. : (B)]

DIBY-52

$\tan^6\frac{\pi}{9}-33\tan^4\frac{\pi}{9}+27\tan^2\frac{\pi}{9}$ is equal to

[Ans. : 3]

DIBY-53

If $5\left(\tan^2x-\cos^2x\right)=2\cos2x+9$, then the value of $\cos4x$ is [JEE Mains-2017]

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $-\frac{7}{9}$ (D) $-\frac{3}{5}$

[Ans. : (C)]

DIBY-54

Find the summation of the following series

(i) $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$

(ii) $\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$

[Ans. : (i) -1/2 (ii) 0]

DIBY-55

Prove that $\sin\theta+\sin 3\theta+\sin 5\theta+\cdots+\sin(2n-1)\theta=\frac{\sin^2 n\theta}{\sin\theta}$

DIBY-56

Sum the series : $\sqrt{1+\cos\alpha}+\sqrt{1+\cos 2\alpha}+\sqrt{1+\cos 3\alpha}+\cdots$ **to** n **terms, where** $0<\alpha<\pi$

[Ans. $\sqrt{2}\frac{\sin\frac{n\alpha}{4}}{\sin\frac{\alpha}{4}}\cos\left[(n+1)\frac{\alpha}{4}\right]$**]**

TRIGONOMETRIC SERIES

DIBY-57

Find the value of $\left(\cos^4 1^\circ + \cos^4 2^\circ + \cos^4 3^\circ + \cdots + \cos^4 179^\circ\right) - \left(\sin^4 1^\circ + \sin^4 2^\circ + \sin^4 3^\circ + \cdots + \sin^4 179^\circ\right)$.

[Ans. : -1]

DIBY-58

If $\frac{\sin\theta}{\cos(3\theta)} + \frac{\sin(3\theta)}{\cos(9\theta)} + \frac{\sin(9\theta)}{\cos(27\theta)} + \frac{\sin(27\theta)}{\cos(81\theta)} = \frac{\sin(k\theta)}{2\cos\theta\cos 81\theta}$, then k equals to

[Ans. : 80]

DIBY-59

$\sin\theta \cdot \sec 3\theta + \sin 3\theta \cdot \sec 3^2\theta + \sin 3^2\theta \cdot \sec 3^3\theta + \cdots$ upto n terms $=$

(A) $\frac{1}{2}[\tan 3^n\theta - \tan\theta]$

(B) $\frac{1}{2}[\tan 3^n\theta + \tan\theta]$

(C) $[\tan 3^n\theta + \tan\theta]$

(D) $[\tan 3^n\theta - \tan\theta]$

[Ans. : (A)]

TRIGONOMETRIC SERIES

DIBY-60

If m and n are positive integers satisfying $1+\cos 2\theta+\cos 4\theta+\cos 6\theta+\cos 8\theta+\cos 10\theta=\frac{\cos m\theta \cdot \sin n\theta}{\sin \theta}$, then $m+n$ is equal to

[Ans. : 11]

DIBY-61

Find the sum of series $S=\sin\theta-\sin 2\theta+\sin 3\theta-\sin 4\theta+\cdots+n$ terms.

[Ans. : .....]

DIBY-62

The value of

$P=(\sqrt{3}+\tan 1^{\circ})(\sqrt{3}+\tan 2^{\circ})(\sqrt{3}+\tan 3^{\circ})\ldots(\sqrt{3}+\tan 28^{\circ})(\sqrt{3}+\tan 29^{\circ})$ is

(A) 2^{28} (B) $2^{15}+2-\sqrt{3}$ (C) 2^{29} (D) $2^{14}+2-\sqrt{3}$

[Ans. : (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

- QIBY - Soln.

from 35 to 62
↳ uploaded.

It is never about end result

It's always about Journey

Thank You

#futureIITians